



Herzlich willkommen bei / welcome at
4future

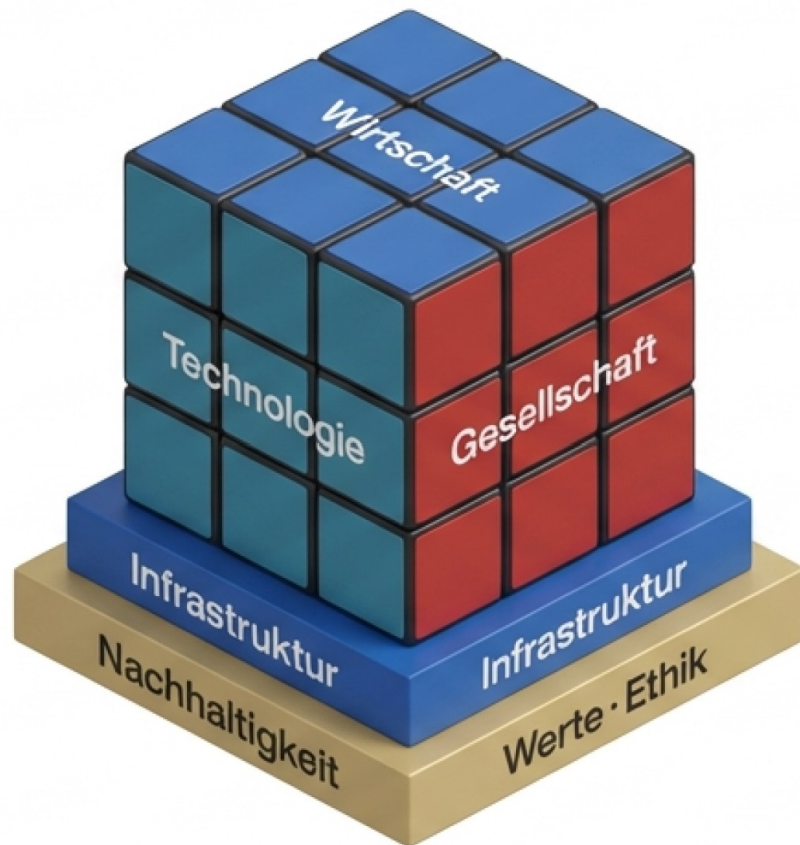


Andi Kunar
andreask@msn.com

Andreas Burner & Andreas Petersson's
Own Your AI

24.6.2025 · 4future space · 1010 Wien

4future ?



Local, non-profit community,
interested in *innovation*

3 views: *.tech*, *.social*, *.business*
with common infrastructure

Independent via volunteering,
donations, and memberships

Online, physical events,
publications, services, ...



4future.tech

Lightning – Talk

AI



Owning your 100GB+ NVIDIA AI-Hardware

... by a predominantly Apple user,
who hates 1500W PSU room-heaters

Andi Kunar
andreask@msn.com



ClubComputer.at
... begleitet Dich in die digitale Zukunft.

NVIDIA DGX Spark



- 128GB unified, cache-coherent RAM @ ~ M4 Pro bandwidth
- Fast Blackwell-series GPU (fp8/fp4, faster than M5 Max / M3 Ultra GPU, thermal limit)
- NVLink 200gbps RDMA scale-out → for 2x RAM/Performance
- It's shipping - e.g. ASUS GX10 @ € 3000 excl. VAT (1TB SSD)
- Linux 24.04 aarch64, CUDA 13, Containers, sm_121 (it's NOT a GB200 GPU)
- Power-consumption: 70 – 200W (it still gets **hot!!!**)
- End of 2026: Windows Version of similar NVIDIA Chips



Running LLMs on the DGX Spark

- Easy (NVIDIA doc) vs. optimized/fast (> 2x faster, sm_121)
- Single node & ease-of-use: llama.cpp (also multimodal!)
- Optimized, experimental: Atlas inference engine, ds4 (for DeepSeek v4)
- Production & clustering: vLLM (complex, spark-vllm-docker)
 - vLLM clustering: tp (splitting width = performance+) or pp (splitting length = throughput+)
- Performance – e.g. 2node cluster (256GB, € 7500) with vLLM
 - MiniMax M2.7 int4 AWQ (max. 150k context) – 2k5 tok/s prefill, 40 tok/s generate
 - DeepSeek V4 fp8 (max. 1M context) – 1k6 tok/s prefill, 40 tok/s generate



Don't forget the NVIDIA Jetsons!

- Embedded systems – robotics, speech, yolo, ...
programmable power envelope, real-time-capable,
Ubuntu 24.04, CUDA 13, unified cache-coherent RAM
- Jetson Orin (Ampere GPU) Nano, 8GB RAM € 300 (excl. VAT)
- Jetson Orin NX16, 16GB RAM € 800
~ M4 Pro GPU, ~ M2 RAM, @ 8-30W Power
- Jetson Orin AGX, 64GB RAM € 2000
- Jetson Thor (Blackwell GPU), 128GB RAM € 3500



My recommendations for “your” big AI



- 128GB RAM, lowest cost: AMD Strix AI Halo System + Fedora
z.B. GMKtec EVO-X2 AMD Ryzen AI Max+ 395, 128GB RAM, 1TB, € 2500 (€ 500 cheaper vs. ASUS GX10, no scale-up)
and use <https://github.com/kyuz0/amd-strix-halo-toolboxes> (Donato Capitella)
- 128GB RAM, agentic performance + expandability: ASUS GX10
or hope for a faster, but more expensive, and without NVLink-expandable M5 Max Mac Studio
- 256GB RAM (MiniMax M2.7, DeepSeek V4): 2x ASUS GX10
a 2x M5 Max Mac Studio cluster with TB5-RDMA can probably only do pp, not tp and likely is slower,
an M3 256GB Ultra Mac Studio is easier, but hard to get, and way more expensive
- >256GB RAM (MiniMax M3,...): hope for a M5 Ultra Mac Studio
... at the price of a car, a 3x DGX Spark cluster is doable (€ 11k) but tricky for parallelism,
a 4x cluster of DGX Sparks also requires an expensive NVLink switch (€ 18k total)

... or rent Datacenter GPUs



M2 Max vs. DGX Spark

M2 Max 96GB MacStudio

lmstudio-community/Qwen3.6-35B-A3B-MLX

lm-studio INT4: pp2048: 715, tg32: 90

mlx-lm INT4: pp2048: 780, tg32: 70

vLLM-metal INT4: pp2048: 655, tg32: 50

DGX-Spark 128GB

Qwen/Qwen3.6-35B-A3B

vLLM FP8: pp2045: 4.900, tg128: 80

vLLM NVFP4: pp2045: 4.000, tg128: 77

Atlas NVFP4: ??? pp2045: 900.000, tg128: 210